Short stories on Bregman divergences: Flat and curved geometries

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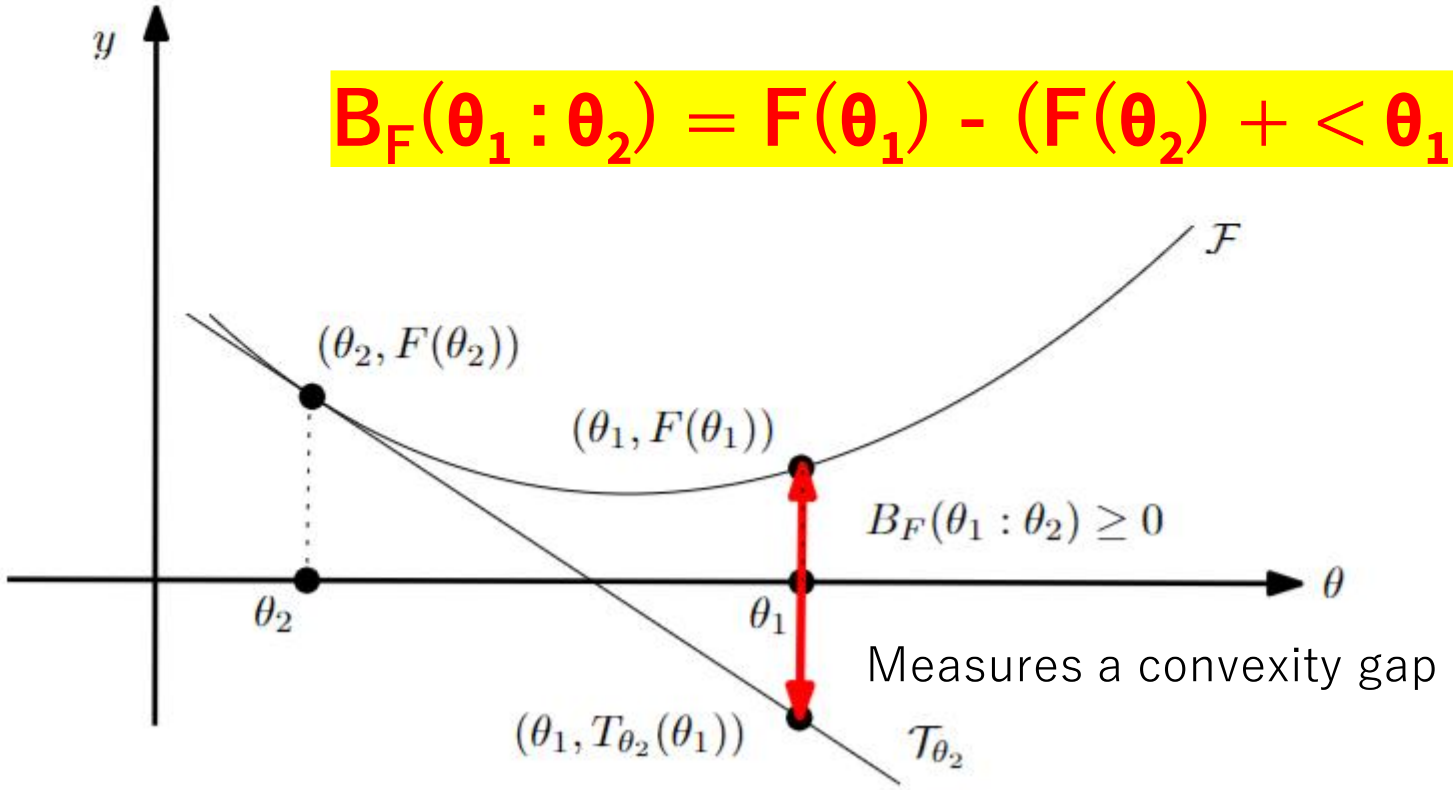
Bregman divergences

- Let $F: \Theta \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ be a strictly convex and smooth real-valued function on a finite-dimensional Hilbert space <.,.>

Bregman divergence $B_F: \Theta \times \mathrm{Int}(\Theta) \rightarrow \mathbb{R}$

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - (F(\theta_2) + \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle)$$



Originally introduced for (left-sided) Bregman projections with applications to LP approximations

A versatile class of dissimilarities

Bregman divergences **unify** various distortions in applications

- ***squared Euclidean divergence*** *from* $F(\theta) = \frac{1}{2}\theta^T Q \theta$, **Quadratic negentropy**:

$$B_F(\theta_1 : \theta_2) = \frac{1}{2}(\theta_2 - \theta_1)^T Q (\theta_2 - \theta_1)^T = \frac{1}{2} ||\theta_2 - \theta_1||^2_Q$$

- ***Kullback-Leibler divergence*** *from* $F(\theta) = \Sigma_i \theta_i \log(\theta_i)$, **Shannon negentropy**:

$$B_F(\theta : \theta') = \Sigma_i \{ \theta_i \log(\theta_i / \theta_i') + \theta_i' - \theta_i \} = \Sigma_i \theta_i \log(\theta_i / \theta_i')$$

- ***Itakura-Saito divergence*** *from* $F(\theta) = \Sigma_i -\log(\theta_i)$, **Burg negentropy**:

$$B_F(\theta : \theta') = \Sigma_i \{ (\theta_i / \theta_i') - \log(\theta_i' / \theta_i) - 1 \}$$

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - (F(\theta_2) + < \theta_1 - \theta_2, \nabla F(\theta_2) >)$$

BD interpreted as **remainder** of a first order Taylor expression of $F(\theta_1)$ around θ_2:

$$F(\theta_1) = F(\theta_2) + < \theta_1 - \theta_2, \nabla F(\theta_2) > + \underbrace{\mathbf{B_F(\theta_1 : \theta_2)}}_{\textbf{Taylor remainder}}$$

Example of remainder:
Lagrange remainder (smooth C^2 generators):
∇^2**F SPD** $\Rightarrow B_F(\theta_1 : \theta_2) \geq 0$

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$$B_F(\theta_1 : \theta_2) = \frac{1}{2} (\theta_2 - \theta_1)^\top \nabla^2 F(\theta) (\theta_2 - \theta_1) \geq 0 , \theta \in [\theta_1 \theta_2]$$

Bregman divergences in machine learning…

- Kullback-Leibler divergence between two probability densities:

$$D_{KL}[p(x):q(x)] = \int p(x) \log (p(x)/q(x))\, d\mu(x)$$

is **difficult to calculate in closed form** because of the integral ∫ …

- But Kullback-Leibler divergence between two probability densities of a **natural exponential family** with densities $p(x|\theta) \propto \exp(\langle x, \theta \rangle)$ amount to a **reverse Bregman divergence** $B_F^{rev}(\theta_1 : \theta_2) := B_F(\theta_2 : \theta_1)$

$$D_{KL}[p(x|\theta_1) : p(x|\theta_2)] = B_F^{rev}(\theta_1 : \theta_2) = B_F(\theta_2 : \theta_1)$$

Bypass the ∫, ∇F in BD easy to calculate! ⇒ Easy calculations of KLDs

Convex duality via Legendre-Fenchel transform

- Legendre-Fenchel transform of a convex function F:

  $F^*(\eta)=\sup_{\theta\in\Theta}\{<\theta,\eta>-F(\theta)\}$

- Problem: some *tricky functions* with gradient map ∇F domain **not convex**…

  Example: $h(\xi_1,\xi_2)=[(\xi_1^2/\xi_2)+\xi_1^2+\xi_2^2]/4$ on upper plane domain $\Xi=(\xi_1,\xi_2)$

- Thus, we consider "**nice convex functions**" = **Legendre-type functions (Θ,F(θ))**

  (i) Θ open, and (ii) $\lim_{\theta\to\partial\Theta}\|\nabla F(\theta)\|=\infty$

Then we get:

❶ **reciprocal gradient maps** $\eta=\nabla F(\theta)$ and $\theta=\nabla F^*(\eta)$, $\nabla F^*=(\nabla F)^{-1}$

❷ conjugation yields (H,$F^*(\eta)$) of Legendre type

❸ biconjugation is an **involution**: $(H,F^*(\eta))^*=(H^*=\Theta,F^{**}=F(\theta))$

- Convex conjugate: $F^*(\eta)=<\nabla F^{-1}(\eta),\eta>-F(\nabla F^{-1}(\eta))$ since $\eta=\nabla F(\theta)$

Fenchel-Young divergences

- Young inequality: $F(\theta_1)+F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$ with equality when $\eta_2 = \nabla F(\theta_1)$
- Build the Fenchel-Young divergence from the inequality: lhs-rhs ≥0

$$\mathbf{Y_{F,F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle} \geq 0$$

- Mixed and dual parameterizations

$$B_F(\theta_1 : \theta_2) = Y_{F,F^*}(\theta_1 : \eta_2) = Y_{F^*,F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

$\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$,

Reconstructing statistical divergences from Bregman divergences with integral generators

BDs ⇒ Statistical divergences

Negentropy of mixture families ⇒ KLD

$$\mathcal{M} := \left\{ m(x;\eta) = \sum_{i=1}^{k-1} \eta_i p_i(x) + (1 - \sum_{i=1}^{k-1} \eta_i) p_0(x) \ : \ \eta_i > 0, \sum_{i=1}^{k-1} \eta_i < 1 \right\}$$

Bregman generator $G(\eta) = -h(m(x;\eta)) = \int_{x\in\mathcal{X}} m(x;\eta)\log m(x;\eta)\mathrm{d}\mu(x)$

$$\eta_i = [\nabla F(\theta)]_i = \int (p_i(x) - p_0(x)) \log m_\theta(x) \mathrm{d}\mu(x) \qquad F^*(\eta) = -\int p_0(x)\log m_\theta(x)\mathrm{d}\mu(x) = h^\times(p_0 : m_\theta)$$

$$\theta_1^\top \eta_2 = \int \sum_i \theta_1(i) p_i \log m_{\theta_2} \mathrm{d}\mu - \sum_i \theta_1(i) p_0 \log m_{\theta_2} \mathrm{d}\mu.$$

$$\begin{aligned}
B_{F,\mathcal{M}}[m_{\theta_1} : m_{\theta_2}] &:= F(\theta_1) + F^*(\eta_2) - \theta_1^\top \eta_2, \\
&= -h(m_{\theta_1}) - \int p_0(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x) - \int \sum_i \theta_1(i) p_i(x) \log m_{\theta_2}(x)\mathrm{d}\mu(x) \\
&\quad + \sum_i \theta_1(i) p_0(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x), \\
&= -h(m_{\theta_1}) - \int ((1 - \sum_i \theta_1(i)) p_0(x) + \sum_i \theta_1(i) p_i(x)) \log m_{\theta_2}(x) \mathrm{d}\mu(x), \\
&= -h(m_{\theta_1}) - \int m_{\theta_1}(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x), \\
&= \int m_{\theta_1}(x) \log \frac{m_{\theta_1}(x)}{m_{\theta_2}(x)} \mathrm{d}\mu(x), \\
&= D_{\mathrm{KL}}[m_{\theta_1} : m_{\theta_2}].
\end{aligned}$$

Cumulant of exponential families ⇒ KLD*

$$F(\theta) = F_{\mathcal{E}}(p_\theta) = \log\left(\int \exp(\theta^\top t(x))\mathrm{d}\mu(x)\right) \qquad F^*(\eta) = -h(p_\theta) = \int p(x)\log p(x)\mathrm{d}\mu(x)$$

$$\eta_2(i) = E_{p_{\theta_2}}[t_i(x)]$$

$$\sum_i \theta_1(i)\eta_2(i) = E_{p_{\theta_2}}\left[\sum_i \theta_1(i)t_i(x)\right]$$

$$\sum_i \theta_1(i)t_i(x) = (\log p_{\theta_1}(x)) + F(\theta_1)$$

hence $\theta_1^\top\eta_2 = E_{p_{\theta_2}}[\log p_{\theta_1} + F(\theta_1)] = F(\theta_1) + E_{p_{\theta_2}}[\log p_{\theta_1}]$

$$\begin{aligned} B_{F,\mathcal{E}}[p_{\theta_1} : p_{\theta_2}] &= F(\theta_1) + F^*(\eta_2) - \theta_1^\top\eta_2, \\ &= \cancel{F(\theta_1)} - h(p_{\theta_2}) - E_{p_{\theta_2}}[\log p_{\theta_1}] - \cancel{F(\theta_1)} \\ &= E_{p_{\theta_2}}\left[\log\frac{p_{\theta_2}}{p_{\theta_1}}\right] =: D_{\mathrm{KL}^*}[p_{\theta_1} : p_{\theta_2}], \end{aligned}$$

BD induced by cumulant function = **reverse KLD** on corresponding densities

Partition function of exponential families ⇒ eKLD*

- The **partition function** Z=exp(F) is **log-convex** and hence also convex. We get two BDs B_Z and B_F from exp. families.

$$
\begin{aligned}
D_{\mathrm{KL}}(\tilde{p}_{\theta_1} : \tilde{p}_{\theta_2}) &= \int \left(\tilde{p}_{\theta_1}(x) \log \frac{\tilde{p}_{\theta_1}(x)}{\tilde{p}_{\theta_2}(x)} + \tilde{p}_{\theta_2}(x) - \tilde{p}_{\theta_1}(x) \right) \mathrm{d}\mu(x), \\
&= \int \left(e^{\langle t(x), \theta_1 \rangle} \langle \theta_1 - \theta_2, t(x) \rangle + e^{\langle t(x), \theta_2 \rangle} - e^{\langle t(x), \theta_1 \rangle} \right) \mathrm{d}\mu(x), \\
&= \left\langle \int t(x) e^{\langle t(x), \theta_1 \rangle} \mathrm{d}\mu(x), \theta_1 - \theta_2 \right\rangle + Z(\theta_2) - Z(\theta_1), \\
&= \langle \theta_1 - \theta_2, \nabla Z(\theta_1) \rangle + Z(\theta_2) - Z(\theta_1) = B_Z(\theta_2 : \theta_1),
\end{aligned}
$$

$$p_\lambda(x) \propto \tilde{p}_\lambda(x) = \exp\left(\langle \theta(\lambda), t(x) \rangle\right) h(x)$$

$$p_\lambda(x) = \frac{1}{Z(\lambda)} \tilde{p}_\lambda(x)$$

$$Z(\lambda) = \int \tilde{p}_\lambda(x) \mathrm{d}\mu(x)$$

2312.12849

Sub-dimensional families of Bregman divergences

Proposition *A multivariate Bregman divergence* $B_F(\theta:\theta')$ *can be written equivalently as a scalar Bregman divergence* $B_{F_{\theta,\theta'}}(0:1)$ *so that*

$$\forall \theta \in \Theta, \theta' \in \mathrm{int}(\theta), \quad B_F(\theta:\theta') = B_{F_{\theta,\theta'}}(0:1),$$

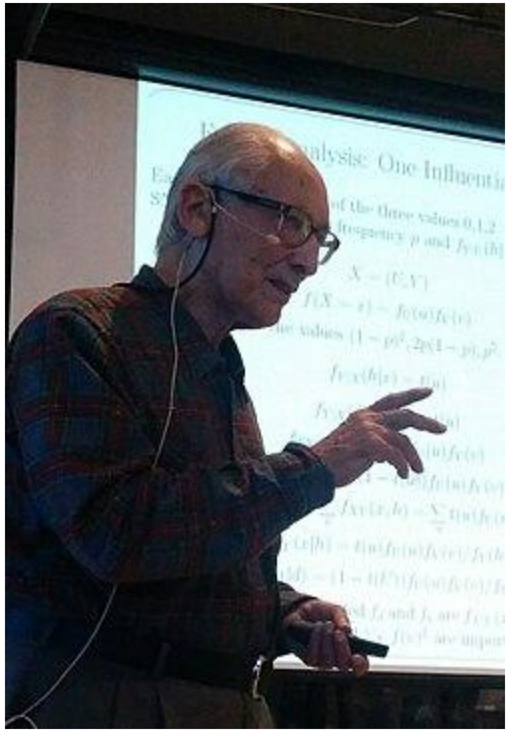
where $F_{\theta,\theta'}(u) := F(\theta + u(\theta'-\theta))$ *is a univariate Bregman generator.*

$$\begin{aligned} B_{F_{\theta,\theta'}}(0:1) &:= F_{\theta,\theta'}(0) - F_{\theta,\theta'}(1) - (0-1)\nabla_{\theta'-\theta} F_{\theta,\theta'}(1), \\ &= F(\theta) - F(\theta') + \langle \theta'-\theta), \nabla F(\theta') \rangle = B_F(\theta:\theta'). \end{aligned}$$

$$\begin{aligned} \nabla_{\theta'-\theta} F_{\theta,\theta'}(u) &:= \lim_{\epsilon\rightarrow 0} \frac{F(\theta+(\epsilon+u)(\theta'-\theta)) - F(\theta+u(\theta'-\theta))}{\epsilon}, \\ &= \langle \theta'-\theta, \nabla F(\theta+u(\theta'-\theta)) \rangle. \end{aligned}$$

Information geometry of Chernoff information

Chernoff distribution, Chernoff point, LREF



July 1, 1923 (age 103)

$$C(P,Q) = -\log \min_{\alpha\in(0,1)} \int p^{\alpha}(x) q^{1-\alpha}(x) \mathrm{d}\nu(x)$$

CI useful in statistics, information fusion, etc.

$$C(P_1, P_2) \stackrel{\triangle}{=} -\min_{0 \le \lambda \le 1} \log \left(\sum_x P_1^{\lambda}(x) P_2^{1-\lambda}(x) \right)$$

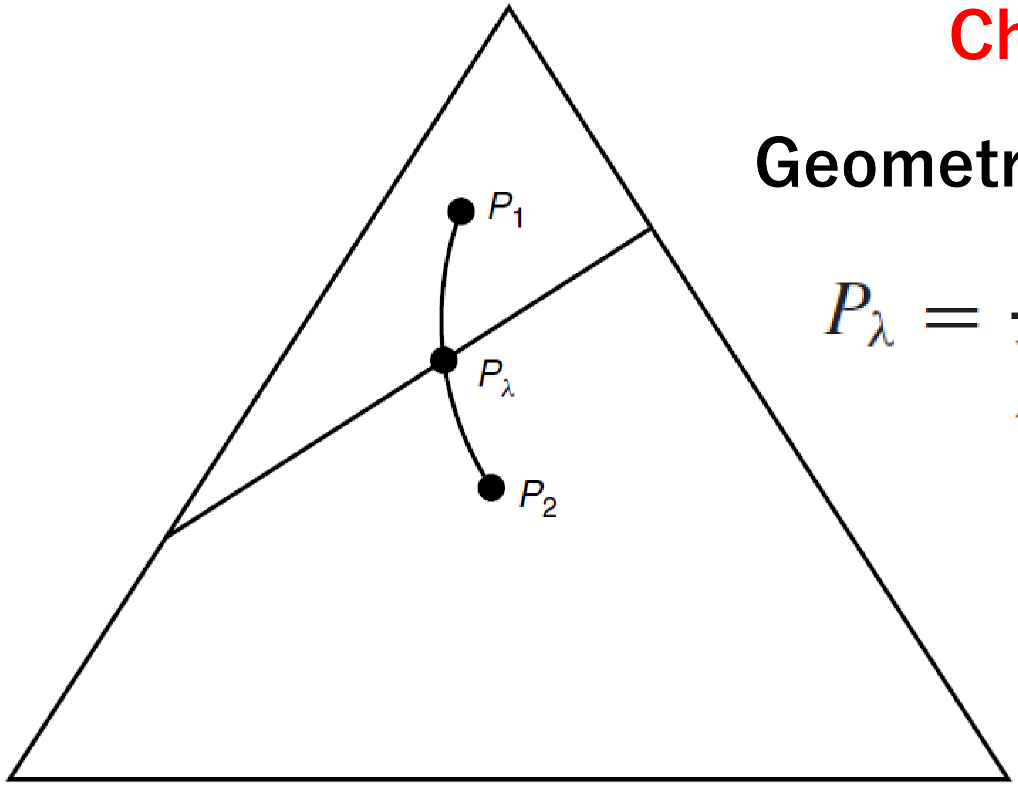
$$= D(P_{\lambda^*} || P_1) = D(P_{\lambda^*} || P_2)$$

Voronoi bisector

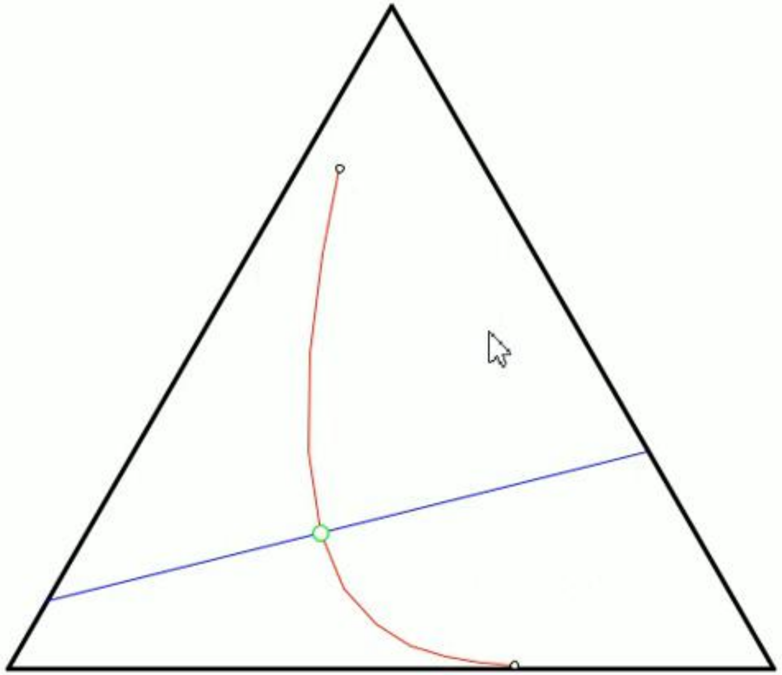
Chernoff distribution

Geometric mixture

$$P_\lambda = \frac{P_1^{\lambda}(x) P_2^{1-\lambda}(x)}{\sum_{a \in \mathcal{X}} P_1^{\lambda}(a) P_2^{1-\lambda}(a)}$$



(Fig. from Thomas & Cover textbook)



Likelihood ratio exponential families (LREFs)

exponential open arc $\{g_\alpha(x) \;:\; \alpha \in (0,1)\}$

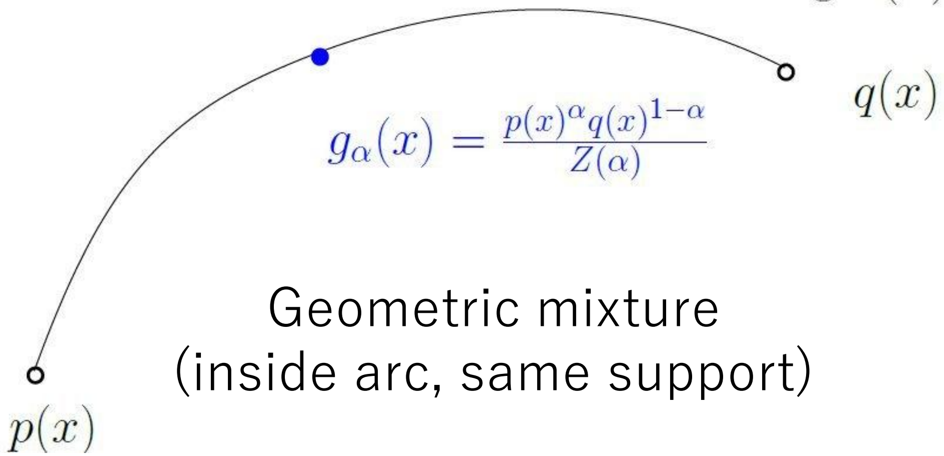
is an exponential family $g_\alpha(x) = \exp\left(\alpha \log \frac{p(x)}{q(x)} - \log Z(\alpha)\right) q(x)$

$Z(\alpha) = \int p(x)^\alpha q(x)^{1-\alpha} \mathrm{d}\mu(x)$

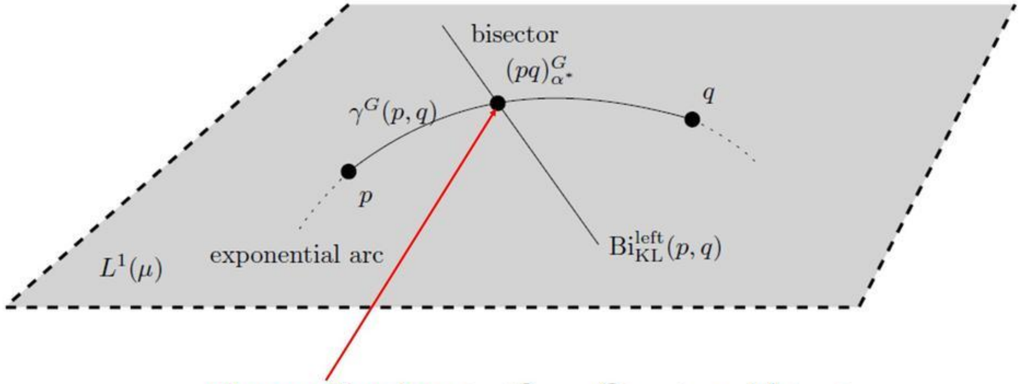
Convex

$\log Z(\alpha) = -D_\alpha^{\mathrm{Bhattacharyya}}[p : q]$

Cumulant of LREF is negative Bhatt. dist.



Geometric mixture
(inside arc, same support)



Chernoff point $(pq)^G_{\alpha^*} = \gamma^G(p,q) \cap \mathrm{Bi}^{\mathrm{left}}_{\mathrm{KL}}(p,q)$

$\gamma^G(p,q) := \left\{(pq)^G_\alpha \;:\; \alpha \in [0,1]\right\}$

$\mathrm{Bi}^{\mathrm{left}}_{\mathrm{KL}}(p,q) := \left\{r \in L^1(\mu) \;:\; D_{\mathrm{KL}}[r:p] = D_{\mathrm{KL}}[r:q]\right\}$

Bhattacharya distance is not a metric distance

$$D_{\mathrm{Bhat},\alpha}[p:q] := -\log \int_{\mathcal{X}} p(x)^\alpha q(x)^{1-\alpha} \,\mathrm{d}\mu(x)$$ Concave in α

"Revisiting Chernoff information with likelihood ratio exponential families." Entropy（2022）

Estimating KLDs/f-divergences

extended to positive densities

Monte Carlo estimation of ext. KL/f-divergences

$$D_{\mathrm{KL}}(p:q) = \int p(x) \log \frac{p(x)}{q(x)} \mathrm{d}\mu(x) \xrightarrow[\text{sampled from p}]{x_1, \cdots, x_n} \widehat{\mathrm{KL}}_n(p:q) = \frac{1}{n} \sum_{i=1}^{n} \log \frac{p(x_i)}{q(x_i)}$$

Problem: Estimated KLD may be negative. Bug/error in practice!

Definition 1 (Extended f-divergence) *The extended f-divergence for a convex generator f, strictly convex at 1 and satisfying $f(1) = 0$ is defined by*

$$I_f^e(p:q) = \int p(x) \left(f\left(\frac{q(x)}{p(x)}\right) - f'(1)\left(\frac{q(x)}{p(x)} - 1\right)\right) \mathrm{d}\mu(x).$$

$$= \int p(x) \underbrace{B_f\left(\frac{q(x)}{p(x)} : 1\right)}_{\geq 0} \mathrm{d}\mu(x) \geq 0.$$

f-divergences are equivalent modulo affine terms

$$\widehat{\mathrm{KL}}_n(p:q) = \frac{1}{n} \sum_{i=1}^{n} \left(\log \frac{p(x_i)}{q(x_i)} + \frac{q(x_i)}{p(x_i)} - 1 \right) \geq 0$$

Non-negative Monte Carlo estimation of f-divergences, 2020

Sigma points for the KLD between EF densities

- Bregman divergences $B_F = B_G$ iff $F \equiv G$ modulo an **affine term**
- Thus consider the equivalent generator $\boxed{\mathrm{F}_\omega(\theta) := -\log(\mathrm{p}_\theta(\omega))}$ for any $\omega \in \mathcal{X}$:

$$p_\theta(x) = \exp(\theta^\top t(x) - F(\theta))$$

$$F_\omega(\theta) := -\log(p_\theta(\omega)) = F(\theta) - \underbrace{(\theta^\top t(\omega) + k(\omega))}_{\text{affine term in } \theta}$$

- Since we have $\nabla F_\omega(\theta(\lambda_1)) = -(t(\omega) - \nabla F(\theta(\lambda_1)))$:

$$\begin{aligned} D_{\mathrm{KL}}[p_{\lambda_1} : p_{\lambda_2}] &= B_{F_\omega}(\theta(\lambda_2) : \theta(\lambda_1)), \\ &= \log\left(\frac{p_{\lambda_1}(\omega)}{p_{\lambda_2}(\omega)}\right) - (\theta(\lambda_2) - \theta(\lambda_1))^\top \nabla_\theta F_\omega(\theta(\lambda_1)) \end{aligned}$$

$$\boxed{D_{\mathrm{KL}}[p_{\lambda_1} : p_{\lambda_2}] = \log\left(\frac{p_{\lambda_1}(\omega)}{p_{\lambda_2}(\omega)}\right) + (\theta(\lambda_2) - \theta(\lambda_1))^\top (t(\omega) - \nabla F(\theta(\lambda_1))), \quad \forall \omega \in \mathcal{X}}$$

KLD for multi-order exponential families

- Choose $s \leqslant D+1$ values for ω (i.e., $\omega_1, \ldots, \omega_s$), and **average**:

$$\boxed{D_{\mathrm{KL}}[p_{\lambda_1} : p_{\lambda_2}] = \frac{1}{s}\sum_{i=1}^{s} \log\left(\frac{p_{\lambda_1}(\omega_i)}{p_{\lambda_2}(\omega_i)}\right) + (\theta(\lambda_2)-\theta(\lambda_1))^\top \left(\frac{1}{s}\sum_{i=1}^{s} t(\omega_i) - \nabla F(\theta(\lambda_1))\right)}$$

Theorem (KLD as sum of log density-ratios)

The Kullback-Leibler divergence between two densities p_{λ_1} and p_{λ_2} belonging to a full regular exponential family $\mathcal{E}$ of order D can be expressed as the average sum of logarithms of density ratios:

$$\boxed{D_{\mathrm{KL}}[p_{\lambda_1} : p_{\lambda_2}] = \frac{1}{s}\sum_{i=1}^{s} \log\left(\frac{p_{\lambda_1}(\omega_i)}{p_{\lambda_2}(\omega_i)}\right) \text{ such that } \frac{1}{s}\sum_{i=1}^{s} t(\omega_i) = E_{p_{\lambda_1}}[t(x)]}$$

where $\omega_1, \ldots, \omega_s$ are $s \leqslant D+1$ distinct points of $\mathcal{X}$ chosen such that $\frac{1}{s}\sum_{i=1}^{s} t(\omega_i) = E_{p_{\lambda_1}}[t(x)]$.

Example: KLD between two multivariate normals (MVNs)

- Exponential family of d-dimensional no[illegible]sities: **order** $D = \frac{d(d+3)}{2}$.
- Use $s = 2d \leqslant D + 1$ **vectors** ω_i to express [illegible] $\Sigma_1 : p_{\mu_2,\Sigma_2}]$:

$$\omega_i = \mu_1 - \sqrt{d\lambda_i} e_i, \omega_{i+d} = \mu_1 + \sqrt{d\lambda_i} e_i, \text{[illegible]} 1, \ldots, d\}$$

where the λ_i's are the **eigenvalues** of Σ_1 with the e_i's [illegible]esponding **eigenvectors**.

- We have $\frac{1}{2d}\sum_{i=1}^{2d}\omega_i = E_{p_{\lambda_1}}[x] = \mu_1$ and $\frac{1}{2d}\sum_{i=1}^{2d}\omega_i\omega_i^\top = \mu_1\mu_1$[illegible]
- Let $\sqrt{\lambda_i}e_i = [\sqrt{\Sigma_1}]_{\cdot,i}$, the i-th column of the **square root of the covariance matrix** of Σ_1. We have:

$$D_{\mathrm{KL}}[p_{\mu_1,\Sigma_1} : p_{\mu_2,\Sigma_2}] =$$

$$\frac{1}{2d}\sum_{i=1}^{d}\left(\log\left(\frac{p_{\mu_1,\Sigma_1}(\mu_1 - [\sqrt{d\Sigma_1}]_{\cdot,i})}{p_{\mu_2,\Sigma_2}(\mu_1 - [\sqrt{d\Sigma_1}]_{\cdot,i})}\right) + \log\left(\frac{p_{\mu_1,\Sigma_1}(\mu_1 + [\sqrt{d\Sigma_1}]_{\cdot,i})}{p_{\mu_2,\Sigma_2}(\mu_1 + [\sqrt{d\Sigma_1}]_{\cdot,i})}\right)\right)$$

where $[\sqrt{d\Sigma_1}]_{\cdot,i} = \sqrt{\lambda_i}e_i$ denotes the vector extracted from the i-th column of the square root matrix of $d\Sigma_1$.

Characterizing Monte Carlo estimation error

- Monte Carlo KLD estimation: $\tilde{D}_{\mathrm{KL}}^{\mathcal{S}_m}[p_{\lambda_1} : p_{\lambda_2}] = \frac{1}{m}\sum_{i=1}^{m} \log \frac{p_{\lambda_1}(x_i)}{p_{\lambda_2}(x_i)}$

- The MC error can be **exactly quantified** for EF densities as

$$\left|\tilde{D}_{\mathrm{KL}}^{\mathcal{S}_m}[p_{\lambda_1} : p_{\lambda_2}] - D_{\mathrm{KL}}[p_{\lambda_1} : p_{\lambda_2}]\right| = \left|(\theta(\lambda_2) - \theta(\lambda_1))^\top \left(\frac{1}{m}\sum_{i=1}^{m} t(x_i) - E_{p_{\lambda_1}}[t(x)]\right)\right|$$

When $m \rightarrow \infty$, we have $\frac{1}{m}\sum_{i=1}^{m} t(x_i) \rightarrow E_{p_{\lambda_1}}[t(x)]$

Consistent estimator

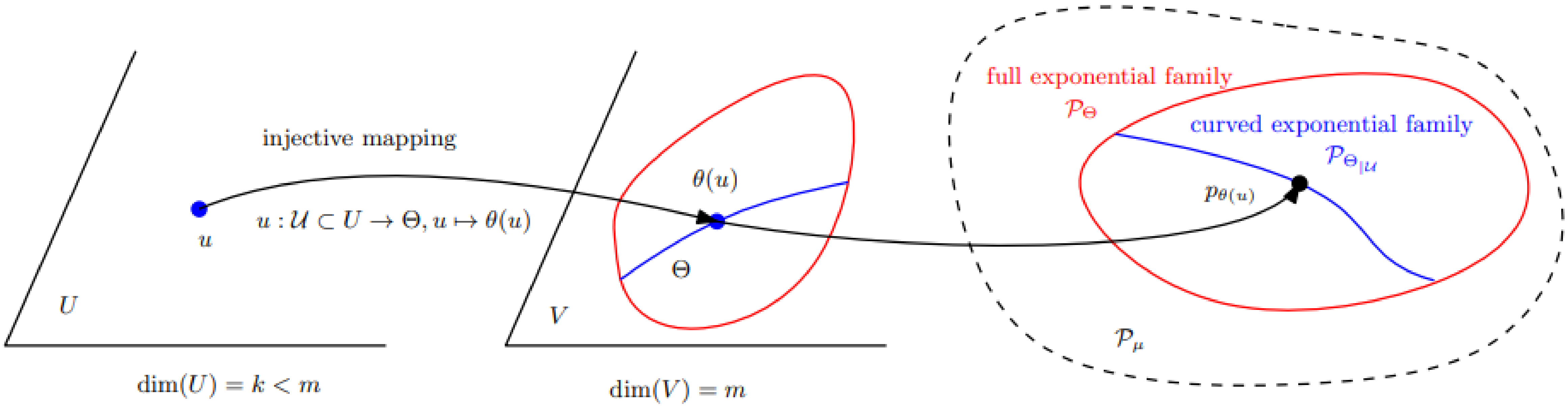
"Computing statistical divergences with sigma points." GSI 2021, Springer LNCS.

Curved Bregman divergences

Curved Bregman divergences (cBDs)

Consider a domain U which maps to a subset of Θ by $\theta = c(u)$ with **dim(U) < dim(Θ)** : $B_{F,c}(u_1 : u_2) := B_F(c(u_1) : c(u_2))$ is not Bregman when $\{c(u) \mid u \in U\}$ not convex. cBDs usually not BDs unless constraints c(.) are affine

By analogy to **curved exponential families**



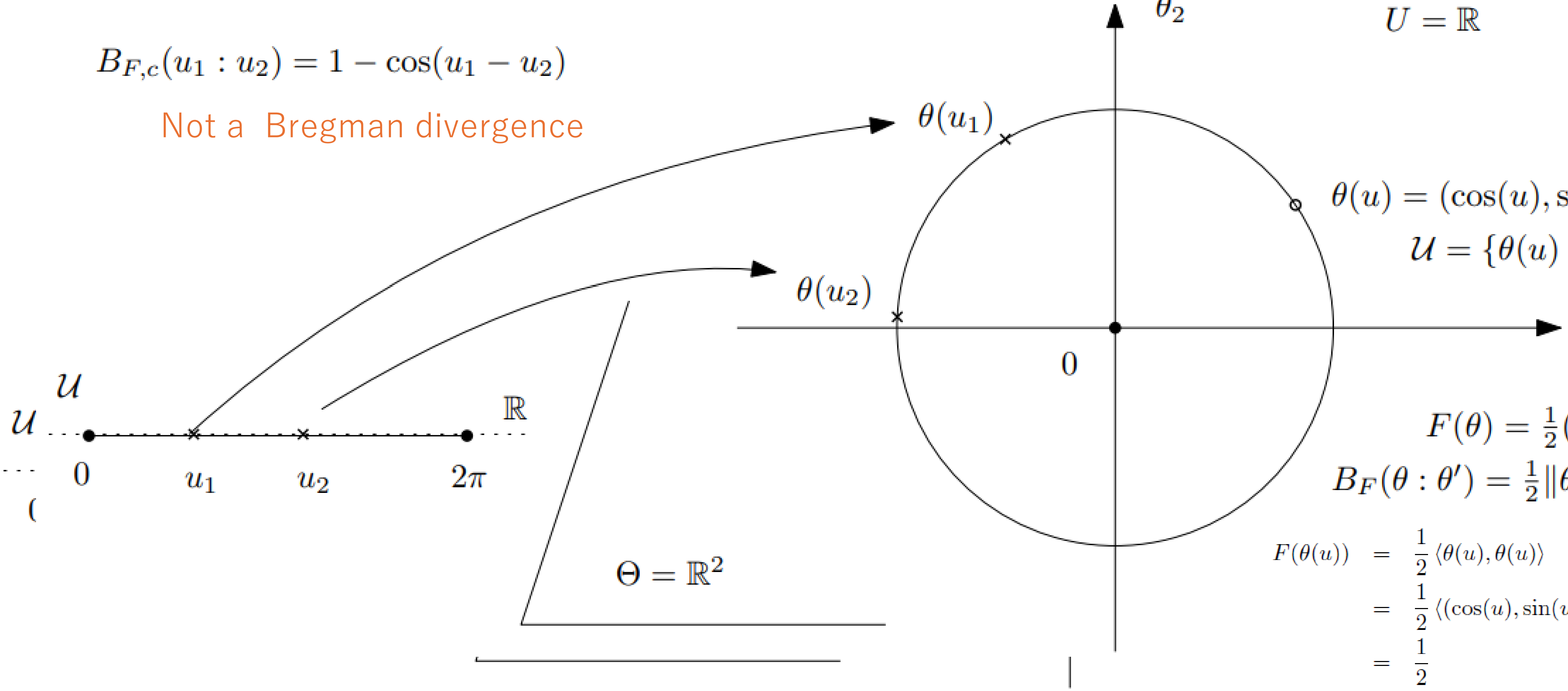
For example, U is real vector space,
V is symmetric matrix vector space Σ(u)
CEF is sub-family of Gaussians { N(0, Σ(u)) : u in U }

Curved Bregman divergence example:
The cosine dissimilarity between unit vectors

Curved Bregman divergence: $B_{F,\theta}(u_1 : u_2) \quad = \quad B_F(\theta(u_1) : \theta(u_2))$

$B_{F,c}(u_1 : u_2) = 1 - \cos(u_1 - u_2)$

Not a Bregman divergence



$U = \mathbb{R}$

$\theta(u) = (\cos(u), \sin(u))$

$\mathcal{U} = \{\theta(u) \ : u \in [0, 2\pi)\}$

$F(\theta) = \frac{1}{2}(\theta_1^2 + \theta_2^2)$

$B_F(\theta : \theta') = \frac{1}{2}\|\theta - \theta'\|^2$

$$\begin{aligned} F(\theta(u)) &= \frac{1}{2}\langle\theta(u), \theta(u)\rangle \\ &= \frac{1}{2}\langle(\cos(u), \sin(u)), (\cos(u), \sin(u))\rangle \\ &= \frac{1}{2} \end{aligned}$$

arXiv:2504.05654

A constant on the circle submanifold, thus not a proper Bregman generator!

Example of curved BDs: Symmetrized BDs

= **Jeffreys-Bregman divergences** are curved Bregman divergences:

$S_F(\theta_1, \theta_2) = \langle \theta_1 - \theta_2, \eta_1 - \eta_2 \rangle$

$$\begin{aligned} S_F(\theta_1 : \theta_2) &= B_F(\theta_1 : \theta_2) + B_F(\theta_2 : \theta_1), \\ &= B_F(\theta_1 : \theta_2) + B_{F^*}(\nabla F(\theta_1) : \nabla F(\theta_2)) \\ &= \breve{B}_{F_\xi}(\xi(\theta_1) : \xi(\theta_2)), \end{aligned}$$

Curved domain: $\mathcal{U} = \{(\theta, \nabla F(\theta)) : \theta \in \Theta\}$

$\xi(\theta) = (\theta, \nabla F(\theta)) \qquad F_\xi(\theta, \eta) := F(\theta) + F^*(\eta) \qquad F^*(\eta) = \langle \theta, \eta \rangle - F(\theta)$

m-dimensional submanifold in 2m-dimensional space
(usually not cvx affine space, hence *not a Bregman divergence*)
Only affine for quadratic generators (i.e., squared Mahalanobis divergences)

"Sided and symmetrized Bregman centroids." IEEE transactions on Information Theory 55.6 (2009)

Another example of curved Bregman divergences:

- Consider d-variate **circular complex normal distributions** $\mathcal{CN}_d(\mu_{\mathbb{C}}, S_{\mathbb{C}})$
- CNDs handled as **2d real normal distributions** $N_{2d}([\mu_{\mathbb{C}}]_{\mathbb{R}}, \frac{1}{2}[S_{\mathbb{C}}]_{\mathbb{R}})$

$$[z = a + ib]_{\mathbb{R}} = (a, b) \qquad [M = A + iB]_{\mathbb{R}} = \left[\begin{array}{cc} A & -B \\ B & A \end{array}\right]$$

- Then KLD between CNDs amounts to a **curved Bregman divergence**:

$$D_{\mathrm{KL}}[p_{m_{\mathbb{C}},S_{\mathbb{C}}} : p_{m'_{\mathbb{C}},S'_{\mathbb{C}}}] = D_{\mathrm{KL}}[p_{\mu,\Sigma} : p_{\mu',\Sigma'}] = B_F(\theta' : \theta)$$

Natural parameters

$(\Sigma^{-1}\mu, \frac{1}{2}\Sigma^{-1})$

$\mu = [m_{\mathbb{C}}]_{\mathbb{R}}$ $\Sigma = [S_{\mathbb{C}}]_{\mathbb{R}}$ and $\mu' = [m'_{\mathbb{C}}]_{\mathbb{R}}$ $\Sigma' = [S'_{\mathbb{C}}]_{\mathbb{R}}$

- **Curved submanifold** parameter in R^{2d} x Mat(2d)

$$\mathcal{U} = \left\{(v, M) : v \in \mathbb{R}^{2d}, M = \left[\begin{array}{cc} A & -B \\ B & A \end{array}\right], A \in \mathbb{R}^{d\times d} \succ 0, B \in \mathbb{R}^{d\times d} \succ 0\right\}$$

2504.05654

See also Nakashika, T: "Complex-Valued Variational Autoencoder: A Novel Deep Generative Model for Direct Representation of Complex Spectra." *INTERSPEECH*. 2020.

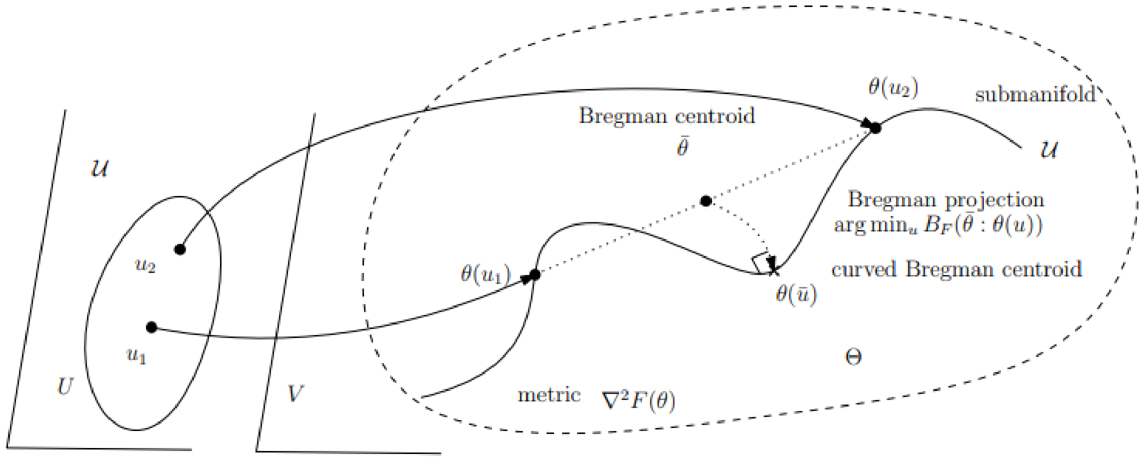
Theorem: Curved Bregman centroid is the Bregman projection of the full Bregman centroid

Right Bregman projection of Bregman centroid $\bar\theta = \sum_i w_i \theta_i$

$$\arg\min_{u\in\mathcal{U}} \sum_{i=1}^n w_i B_F(\theta_i : \theta(u)) = \arg\min_{u\in\mathcal{U}} B_F(\bar\theta : \theta(u))$$

N-point opt. $\theta_i = \theta(u_i)$

From n-point opt to One-point opt!



Simple **Proof.**

$$\begin{aligned}\min_{u\in\mathcal{U}} \sum_{i=1}^n w_i B_F(\theta_i : \theta(u)) &= \sum_{i=1}^n w_i (F(\theta_i) - F(\theta(u)) - \langle \theta_i - \theta(u), \nabla F(\theta(u))\rangle),\\ &\equiv -F(\theta(u)) - \langle \bar\theta - \theta(u), \nabla F(\theta(u))\rangle,\\ &\equiv F(\bar\theta) - F(\theta(u)) - \langle \bar\theta - \theta(u), \nabla F(\theta(u))\rangle\\ &= B_F(\bar\theta : \theta(u)).\end{aligned}$$

"What is... an information projection?" Notices of the AMS 65.3 (2018): 321-324.

Bregman divergences and gauge freedom

Convex functions remain convex by scaling, adding affine term
and reparameterization by affine transformations:
Underlying geometry are dually flat spaces (M,g,∇,∇*).

$$\bar{F}(\theta) = \lambda F(A\theta + b) + \langle c, \theta\rangle + d \Rightarrow \eta = \nabla \bar{F}(\theta) = \lambda A^\top \nabla F(A\theta + b) + c.$$

$$A \in \mathrm{GL}(d, \mathbb{R}), \text{ vectors } b, c \in \mathbb{R}^d \text{ and scalars } d \in \mathbb{R} \text{ and } \lambda \in \mathbb{R}_{>0}$$

$$\nabla \bar{G}(\eta) = A^{-1}\nabla G\left(\frac{1}{\lambda}A^{-\top}(\eta - c)\right) - b \Rightarrow \bar{G}(\eta) = \langle \eta, \nabla \bar{G}(\eta)\rangle - F(\nabla \bar{G}(\eta))$$

$$B_F(\theta_1 : \theta_1) = \frac{1}{\lambda} B_{\bar{F}}(\bar{\theta}_1 : \bar{\theta}_2) \quad \text{with} \quad \bar{\theta} = A^{-1}(\theta - b)$$

$$\boxed{D_{\nabla,\nabla^*}(p_1 : p_2) = B_F(\theta_1 : \theta_1) = \frac{1}{\lambda} B_{\bar{F}}(\bar{\theta}_1 : \bar{\theta}_2)}$$

Affine Legendre Invariance
+
Divergence unit

2301.10980

Hessian (Riemannian) geometry from Bregman generators

Metric tensor in primal coordinates:

$$g_{ij}(\theta) = \nabla^2 F(\theta)$$

Dual metric tensor in dual coordinates:

$$g^{ij}(\eta) = g^{*ij}(\eta) = \nabla^2 F^*(\eta)$$

Crouzeix's identity: $\nabla^2 F(\theta) \nabla^2 F^*(\eta) = I$

A Bregman generator thus define a Riemannian geodesic length distance

Reparameterization of metric tensors $I_\theta(\theta) \xrightarrow{\eta=\eta(\theta)} I_\eta(\eta) = \left[\frac{\partial \theta_i}{\partial \eta_j}\right]^\top \times I_\theta(\theta(\eta)) \times \left[\frac{\partial \theta_i}{\partial \eta_j}\right]$

Separable Bregman divergences （product of 1D EFs…)

… **always** yield **Euclidean geometry** (in non-Cartesian coords)

$$\delta_\Phi(\boldsymbol{x}, \boldsymbol{x}') = \Phi(\boldsymbol{x}) - \Phi(\boldsymbol{x}') - (\boldsymbol{x} - \boldsymbol{x}')^\top \nabla\Phi(\boldsymbol{x}')$$

$$\Phi(\boldsymbol{x}) = \sum_{j=1}^K \phi(x_j) \text{ with } \phi : \mathcal{J} \to \mathbb{R}$$

$$\delta_\Phi(x, x') = \sum_{j=1}^K \delta_\phi(x_j, x'_j)$$

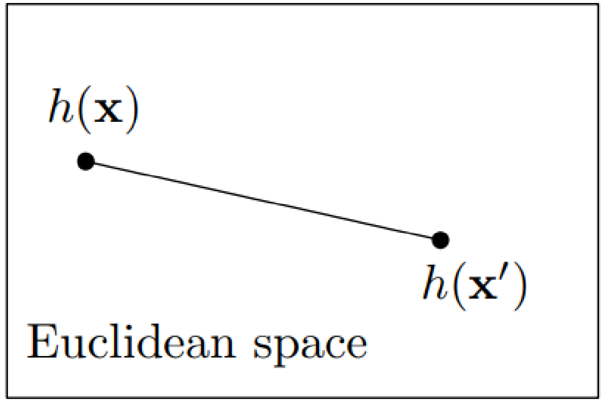
Geodesic distance (Rao distance): $d_\phi(\boldsymbol{x}, \boldsymbol{x}')^2 = \sum_{j=1}^K \left(h(x_j) - h(x'_j)\right)^2$

where $h(x) = \int \sqrt{\phi''(x)}$ $\quad g_{i,j}(\boldsymbol{x}) = \phi''(x_i)\delta_{i,j}$ Diagonal matrix

geodesics $\gamma_i(t) = h^{-1}\Big((1-t)h(x_i) + th(x'_i)\Big)$

$$I_\theta(\theta) \xrightarrow{\eta=\eta(\theta)} I_\eta(\eta) = \left[\frac{\partial\theta_i}{\partial\eta_j}\right]^\top \times I_\theta(\theta(\eta)) \times \left[\frac{\partial\theta_i}{\partial\eta_j}\right]$$

Jacobian matrix y=h(x)



Euclidean dist.
on h-representation

1810.10770

Squaring Hessian matrices yield Euclidean geometry

Crouzeix identity: $$\nabla^2_\eta F^*(\eta)^\top \, \nabla^2 F(\theta(\eta)) = I_{m,m}$$

Squared any Hessian matrix to get **another SPD matrix** representing a metric:

$$G_{\mathrm{sqr}}(\eta) = \nabla^2_\eta F^*(\eta)^\top \, (\nabla^2 F(\theta(\eta)))^2 \nabla^2_\eta F^*(\eta) = I_{m,m}.$$

Geodesic distance is the **Euclidean distance on the dual parameters**:

$$\rho_{g_{\mathrm{sqr}}}(P_1, P_2) = \|\eta_1 - \eta_2\|_2 = \|\nabla F(\theta_1) - \nabla F(\theta_2)\|_2$$

$$\eta_i = \nabla F(\theta(P_i))$$

Instead of Cholesky decomp of SPD, we can take symmetric square root:
It is SPD matrix and square it. Mahalanobis is thus Euclidean geometry 2511.21173

Six short stories

- Bregman divergences for cumulant & partition functions yield reverse KLD and extended reverse KLD divergences
- Log ratio exponential family are 1D exponential family (e-geodesic of geometric mixtures). Application: Chernoff information
- Bregman generators can be chosen modulo affine terms: Characterize MC error of KLD estimation exactly using sigma points
- Curved Bregman centroids are projections of Bregman centroids (symmetrized BDs)
- Bregman divergences and gauge freedom
- Hessian geometry from Bregman generators and Euclidean geometry

Implementing information geometry

Stochastic Bregman generator / Random information geometry
Numerical information geometry / Neural information geometry

Stochastic/random information geometry

- In many cases, dual structures of information geometry are not available in closed-forms or comp. tractable (polynomial EF, continuous mixture family). **With high probability**, we can sample the generator integrals and get proper Bregman generators

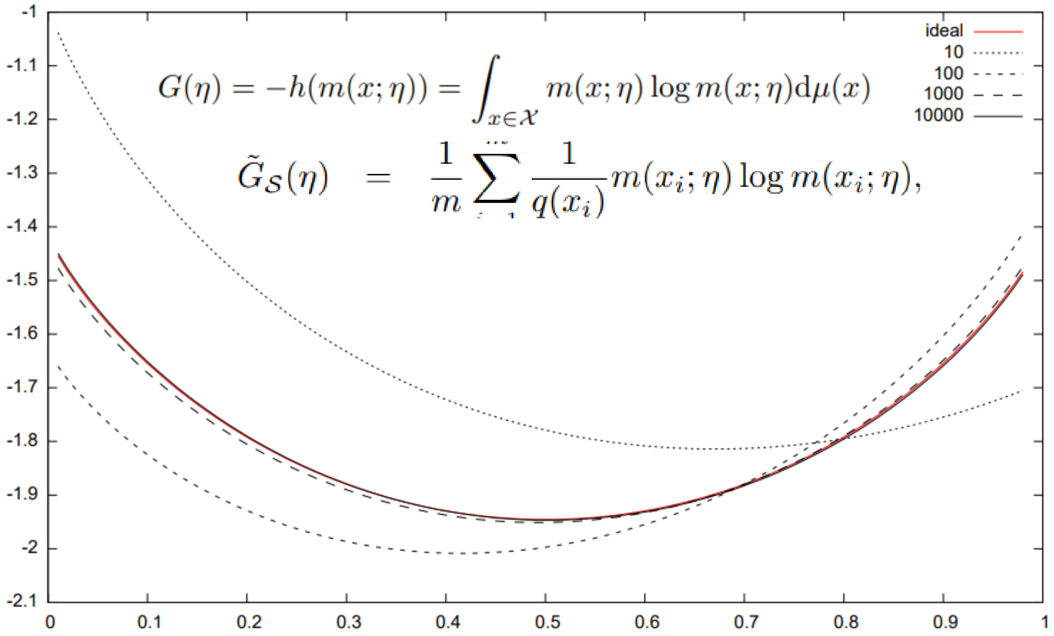
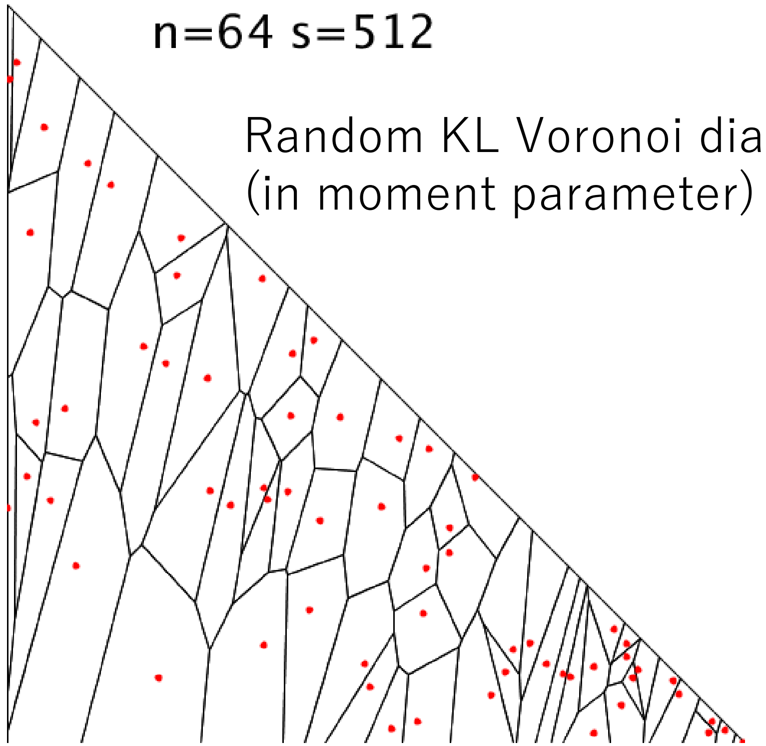


Figure 2: A series $G_{\mathcal{S}}(\eta)$ of Bregman Monte Carlo Mixture Family generators (for $m = |\mathcal{S}| \in \{10, 100, 1000, 10000\}$) approximating the untractable ideal negentropy generator $G(\eta) = -h(m(x;\eta))$ (red) of a mixture family with prescribed Gaussian distributions $m(x;\eta) = (1-\eta)p(x;0,3) + \eta p(x;2,1)$ for the proposal distribution $q(x) = m(x;\frac{1}{2})$.

1803.07225

n=64 s=512



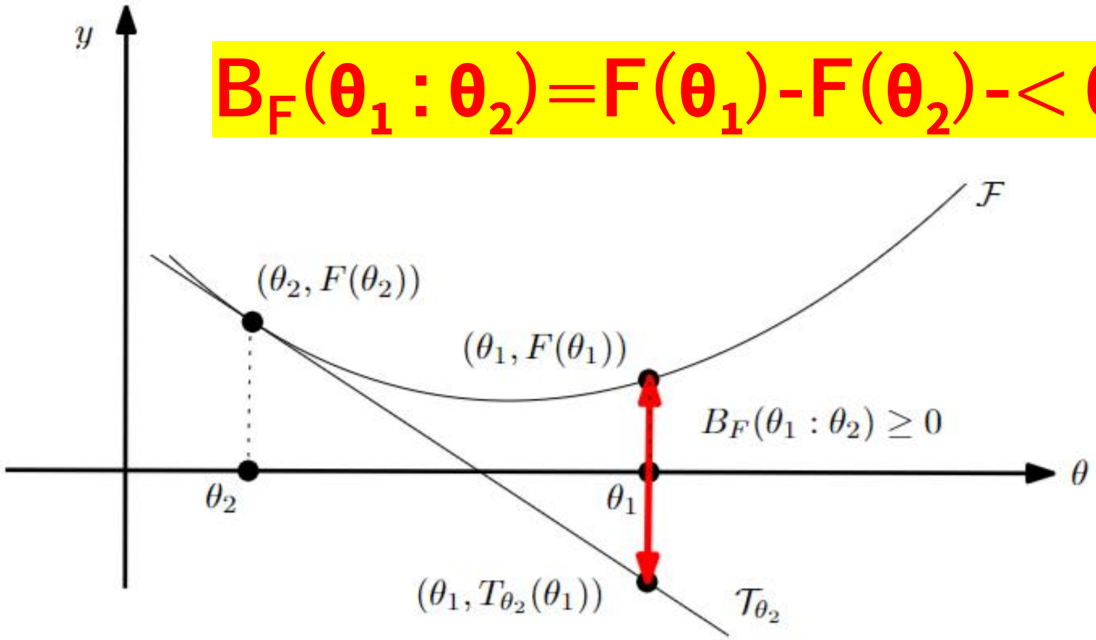
Random KL Voronoi diagram
(in moment parameter)

Dual Bregman/Fenchel-Young divergences

- Let F: $\Theta \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ be a strictly convex and smooth real-valued function on a Hilbert space <.,.>

Bregman divergence $B_F: \Theta \times \mathrm{Int}(\Theta) \rightarrow \mathbb{R}$

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle$$



Popular in geometry, information theory, signal/sound processing!

$$B_F(\theta_1 : \theta_2) = Y_{F,F^*}(\theta_1 : \eta_2) = Y_{F^*,F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Reconstructing statistical divergences from Bregman divergences with integral generators

$$F(\theta) = F_{\mathcal{E}}(p_\theta) = \log\left(\int \exp(\theta^\top t(x))\mathrm{d}\mu(x)\right)$$

$$F_{\mathcal{M}}(m_\theta) = -h(m_\theta) = \int m_\theta(x)\log m_\theta(x)\mathrm{d}\mu(x)$$

Statistical models with dual e/m flatness

- **Exponential families**: $\mathcal{E}_{t,\mu} := \{p(x;\theta) \propto \exp(\langle t(x), \theta \rangle)\}_\theta,$

$$\begin{aligned} \mathrm{KL}(p(x;\theta_1) : p(x;\theta_2)) &= B_F(\theta_2 : \theta_1), \\ &= B_{F^*}(\eta_1 : \eta_2), \end{aligned}$$

$$F(\theta) := \log\left(\int_{x\in\mathcal{X}} \exp(\langle t(x), \theta\rangle) \mathrm{d}\mu(x)\right)$$

$$\eta = E_{p(x;\theta)}[t(x)]$$

FIM $g_{ij}(\theta) = \partial_i \partial_j F(\theta)$

$$g = \mathrm{Var}[t(X)]$$

- **Mixture families**: $\mathcal{M} := \left\{ m(x;\eta) = \sum_{i=1}^{k-1} \eta_i p_i(x) + (1 - \sum_{i=1}^{k-1} \eta_i) p_0(x) \,:\, \eta_i > 0, \sum_{i=1}^{k-1} \eta_i < 1 \right\}$

$$\mathrm{KL}(m(x;\eta) : m(x;\eta')) = B_G(\eta : \eta') \qquad G(\eta) = -h(m(x;\eta)) = \int_{x\in\mathcal{X}} m(x;\eta) \log m(x;\eta) \mathrm{d}\mu(x)$$

FIM $g_{ij}(\eta) = -\partial_i \partial_j h(\eta)$

Remarks:

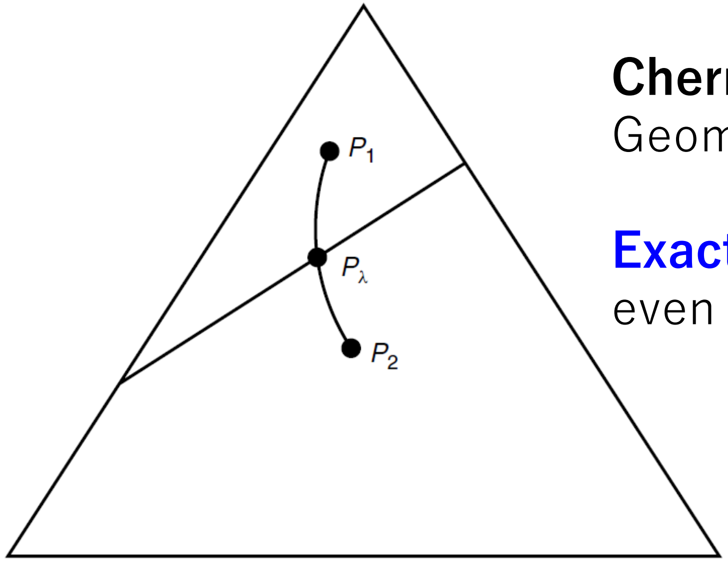
- EFs and MFs are *not flat* with respect to the Levi-Civita g-connection
- Many statistical models are not e/m flat: Ex. Cauchy family

1803.07225

Chernoff information (Bayesian hypothesis testing)

$$C(P_1, P_2) \triangleq -\min_{0\leq\lambda\leq 1} \log\left(\sum_x P_1^\lambda(x) P_2^{1-\lambda}(x)\right)$$
$$= D(P_{\lambda^*}||P_1) = D(P_{\lambda^*}||P_2)$$

$$C(P, Q) = -\log \min_{\alpha\in(0,1)} \int p^\alpha(x) q^{1-\alpha}(x) \mathrm{d}\nu(x).$$



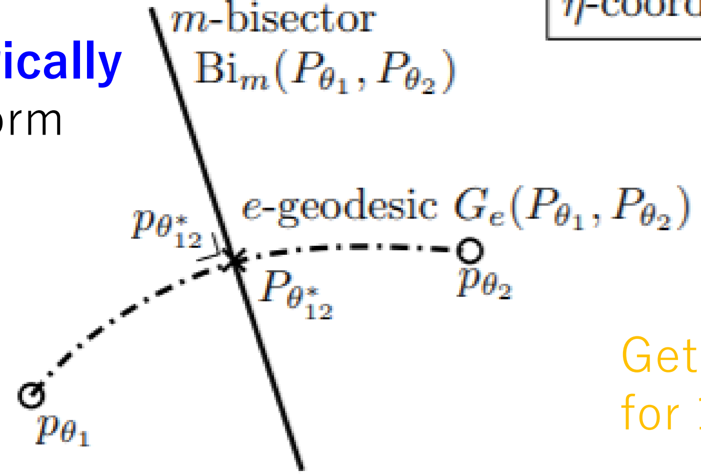
$$P_\lambda = \frac{P_1^\lambda(x) P_2^{1-\lambda}(x)}{\sum_{a\in\mathcal{X}} P_1^\lambda(a) P_2^{1-\lambda}(a)}$$

Chernoff distribution =
Geometric mixture for optimal exponent

Exactly characterized geometrically
even if not available in closed form

$$\boxed{P^* = P_{\theta_{12}^*} = G_e(P_1, P_2) \cap \mathrm{Bi}_m(P_1, P_2)}$$

$\boxed{\eta\text{-coordinate system}}$



$$C(\theta_1 : \theta_2) = B(\theta_1 : \theta_{12}^*)$$

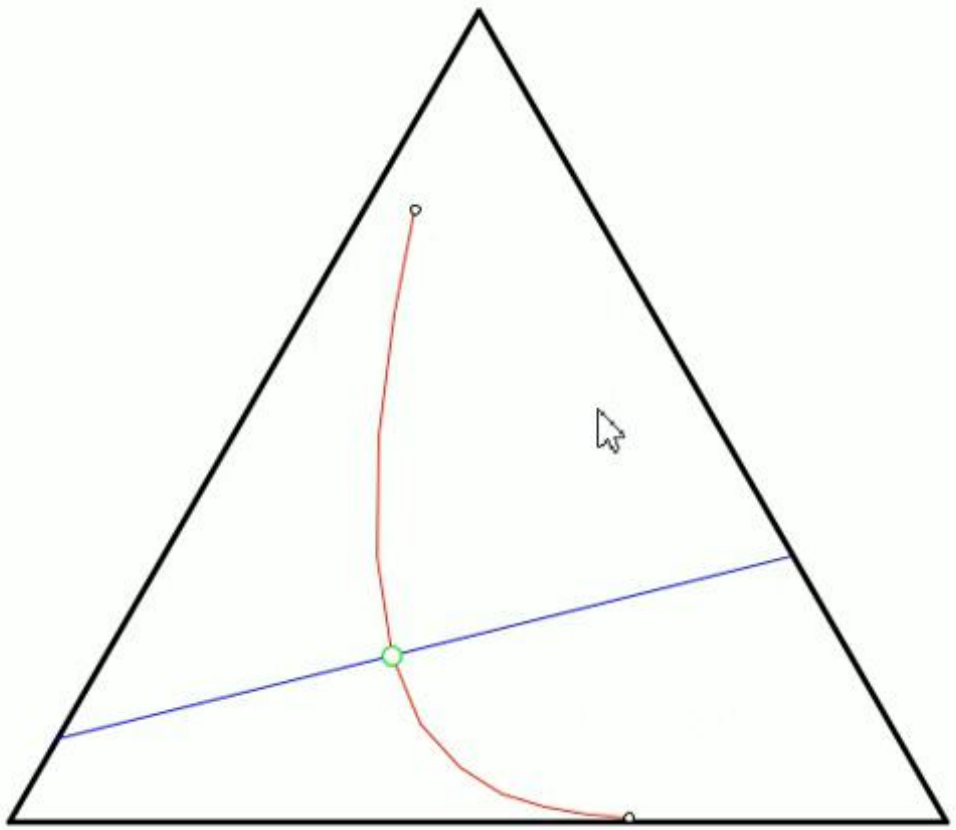
Get large formula
for 1D Gaussians!!!

Categorical/Shannon manifold

Exponential family manifold

In pure geometric term, Chernoff point intersection of primal geodesic with dual bisector

Unique intersection point of
the exponential geodesic
with
the dual mixture bisector



(Here 2D probability simplex of the family of categorical distributions with 3 choices)

Hilbertian metrization of Bregman divergences: Curved Bregman divergences

Bregman divergence for a smooth strictly convex function F

$$B_F = B_F(\theta : \theta') := F(\theta) - F(\theta') - \langle \theta - \theta', \nabla F(\theta') \rangle:$$

Jeffreys symmetrization does not yield square root metrization $\sqrt{S_F}$

$$\begin{aligned} S_F(\theta_1, \theta_2) &:= B_F(\theta_1 : \theta_2) + B_F(\theta_2 : \theta_1) \\ &= \langle \theta_2 - \theta_1, \nabla F(\theta_2) - \nabla F(\theta_1) \rangle \end{aligned}$$

Hilbertian metrization $S_F^{(\alpha,\beta)}(\theta_1, \theta_2) = \|\Phi_{\alpha,\beta}(\theta_1) - \Phi_{\alpha,\beta}(\theta_2)\|_2^2$

$$\Phi_{\alpha,\beta}(\theta) = \begin{bmatrix} \sqrt{\alpha}\theta + \frac{1}{\sqrt{\alpha}} \nabla F(\theta) \\ \sqrt{\frac{\alpha\beta - 1}{\alpha}} \nabla F(\theta) \end{bmatrix}$$

$$\begin{aligned} S_F^{(\alpha,\beta)}(\theta_1, \theta_2) &= \langle \theta_2 - \theta_1, \nabla F(\theta_2) - \nabla F(\theta_1) \rangle \\ &+ \frac{\alpha}{2} \langle \theta_2 - \theta_1, \theta_2 - \theta_1 \rangle + \frac{\beta}{2} \langle \nabla F(\theta_2) - \nabla F(\theta_1), \nabla F(\theta_2) - \nabla F(\theta_1) \rangle. \end{aligned}$$

Hilbertian metrization is a **curved Bregman divergence**

$$\begin{aligned} S_F^{(\alpha,\beta)}(\theta_1, \theta_2) &= \|\Phi_{\alpha,\beta}(\theta_1) - \Phi_{\alpha,\beta}(\theta_2)\|_2^2 \\ &= B_{\hat{F}}(\xi(\theta_1) : \xi(\theta_2)), \end{aligned} \quad \xi(\theta) = (\theta, \nabla F(\theta))$$

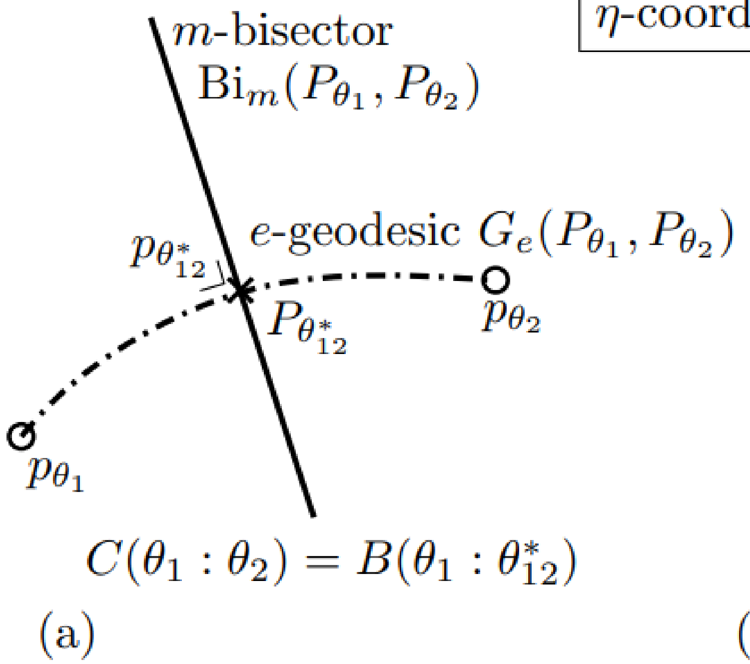
$$\hat{F}_{\alpha,\beta}(\xi_1, \xi_2) = F(\xi_1) + \frac{\alpha}{2} \langle \xi_1, \xi_1 \rangle + F^*(\xi_2) + \frac{\beta}{2} \langle \xi_2, \xi_2 \rangle$$

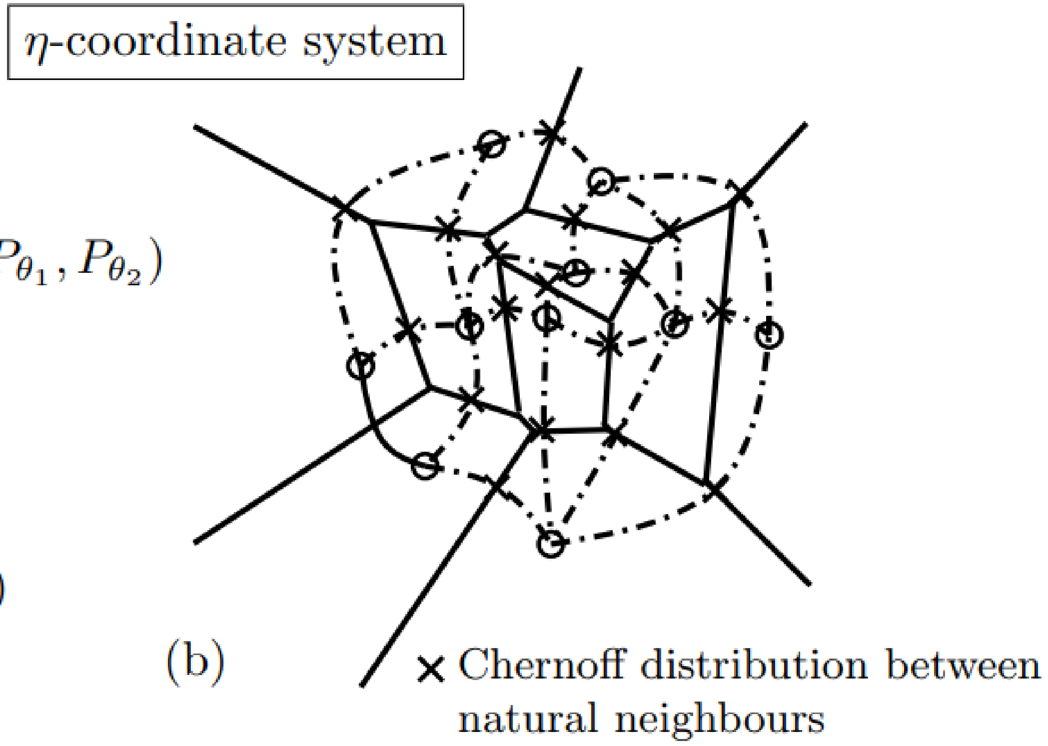
arXiv:2504.05654

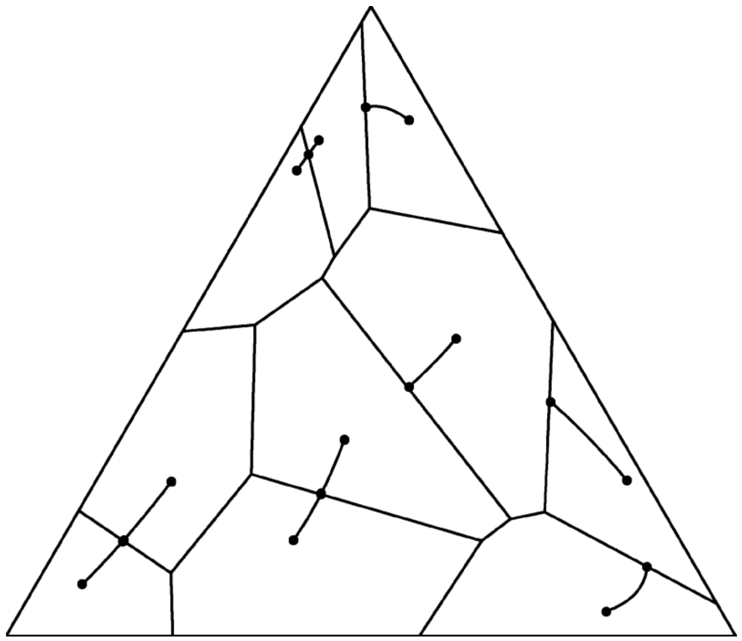
Chernoff information of multiple distributions

Defined as the **minimum of pairwise Chernoff information**

Computed by inspecting **natural neighbors** in **Voronoi diagrams**







Westover, 2008

Bregman Voronoi diagrams

1102.2684

0709.2196